

Spectral Collapse Without Collapse: SVD Dynamics of LoRA Adapters for Low-Resource Turkic Languages

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Abstract

Spectral collapse—the concentration of adapter weight matrices into a single dominant singular direction—has been proposed as a diagnostic for LoRA training failure. We systematically test this hypothesis by fine-tuning Gemma-2-9B with LoRA on three low-resource Turkic languages (Kyrgyz, Kazakh, Uzbek) and an English control, across eight experimental configurations varying rank ($r \in \{16, 64\}$), learning rate (2×10^{-4} , 5×10^{-4} , 1×10^{-3}), training duration (3–10 epochs), and cross-lingual transfer (Kazakh $\rightarrow$ Kyrgyz).

We report three main findings. First, we **challenge the spectral energy threshold hypothesis**: no experiment reaches $SE > 0.59$ (well below the proposed 0.7 threshold), yet configurations with $r = 64$ suffer complete functional collapse (NER F1 = 0 across all languages, cross-lingual perplexity > 600). Spectral energy actually *decreases* during training as effective rank grows. Second, we identify **Frobenius norm growth** as a more reliable indicator of functional degradation in our setting (Gemma-2-9B, 4-bit quantization): pathological configurations exhibit 15–38 $\times$ norm growth versus only +13% for the best-performing transfer setup. Third, we demonstrate that **sequential Kazakh $\rightarrow$ Kyrgyz transfer** yields +61% NER F1 improvement (0.253 vs. 0.157) while retaining Kazakh knowledge (perplexity 23.49 vs. 47.58 for direct Kyrgyz training), suggesting that related-language transfer is an effective strategy for extremely low-resource Turkic varieties.

We additionally characterize severe subword fragmentation (3.2 $\times$ vs. English) as a compounding factor for all three languages and release our SVD monitoring toolkit for reproducible spectral analysis of LoRA training dynamics.

1 Introduction

Low-Rank Adaptation (Hu et al., 2022) has become the dominant approach for adapting large language models (LLMs) to new domains and languages, particularly when computational resources are limited. By decomposing weight updates into low-rank matrices $\Delta W = BA$, LoRA achieves parameter efficiency while preserving most of the pretrained model’s capabilities. However, the internal dynamics of LoRA weight matrices during training—specifically their spectral properties—remain poorly understood, especially for morphologically complex, low-resource languages.

Recent work has identified *spectral collapse* as a potential failure mode of LoRA fine-tuning, where adapter weight matrices degenerate into rank-one approximations dominated by a single singular value (Biderman et al., 2024; Jang et al., 2024). Biderman et al. (2024) proposed that when the spectral energy (SE)—the ratio of the largest singular value to the sum of all singular values—exceeds a threshold of approximately 0.7, the adapter has effectively collapsed and further training becomes unproductive. However, these findings were established primarily on English-language tasks with relatively large training corpora.

Turkic languages present a particularly challenging testbed for LoRA adaptation. Kyrgyz, Kazakh, and Uzbek are agglutinative languages with rich morphological systems, employ both Cyrillic and Latin scripts, and suffer from severe underrepresentation in multilingual LLM pretraining data. Critically, they exhibit extreme subword fragmentation under standard tokenizers (3.2 $\times$ more tokens per word than English), which forces LoRA adapters to make larger weight modifications

to accommodate poorly-covered morphology.

In this work, we systematically investigate the spectral dynamics of LoRA adapters trained on three Turkic languages, testing two hypotheses:

- **H1 (Spectral Energy Threshold):** Spectral energy exceeding 0.7 predicts functional degradation of LoRA adapters (Biderman et al., 2024).
- **H2 (Cross-Lingual Transfer):** Initializing LoRA adapters from a related-language checkpoint (Kazakh $\rightarrow$ Kyrgyz) slows spectral degradation and improves downstream task performance.

Our contributions are as follows:

1. We **challenge the SE threshold hypothesis** for low-resource LoRA fine-tuning: maximum SE never exceeds 0.59, yet functional collapse occurs through alternative mechanisms (§4.3).
2. We propose **Frobenius norm growth ratio** as a scale-aware diagnostic that reliably separates healthy from pathological training regimes *within underrepresented language settings* (§4.4).
3. We demonstrate the **effectiveness of cross-lingual transfer** between related Turkic languages, achieving +61% NER F1 improvement with retained source-language knowledge (§4.5).
4. We provide a quantitative **characterization of subword fragmentation** for Kyrgyz, Kazakh, and Uzbek (§3.6).
5. We release an open-source **SVD monitoring toolkit** for reproducible spectral analysis of LoRA training.

2 Related Work

LoRA and Parameter-Efficient Fine-Tuning. Low-Rank Adaptation (LoRA) (Hu et al., 2022) parameterizes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. QLoRA (Dettmers et al., 2023) extends this by quantizing the base model to 4-bit precision. Subsequent work has explored rank allocation strategies

(Zhang et al., 2023), training stability (Lialin et al., 2023), and theoretical analyses of low-rank update expressiveness (Malladi et al., 2023).

Spectral Analysis of Neural Network Weights. Martin and Mahoney (2021) demonstrated that well-trained networks exhibit heavy-tailed singular value distributions (“implicit self-regularization”). In the LoRA context, Biderman et al. (2024) identified spectral collapse as a potential diagnostic for training degradation, and Jang et al. (2024) proposed monitoring spectral energy as an early warning system. However, both studies focused on English-language settings with large training corpora.

Low-Resource NLP for Turkic Languages. Turkic languages remain severely underrepresented despite recent benchmarks such as TUMLU (Isbarov et al., 2024). Work on Kazakh NLP has leveraged Wikipedia and web-crawled corpora (Yeshpanov et al., 2024), while Kyrgyz and Uzbek remain among the least-resourced languages. The challenge of tokenizer mismatch for agglutinative languages has been documented (Rust et al., 2021; Ahia et al., 2023).

Catastrophic Forgetting in Fine-Tuning. Catastrophic forgetting (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017) remains a persistent challenge. While LoRA’s low-rank constraint was hypothesized to mitigate forgetting (Biderman et al., 2024), aggressive fine-tuning can still induce significant forgetting for distant languages (Csaki et al., 2024). The relationship between spectral properties and forgetting severity has not been systematically studied for multilingual settings.

3 Experimental Setup

3.1 Base Model

We use Gemma-2-9B (Gemma Team et al., 2024) as our base model, a 9.24 billion parameter decoder-only transformer with 42 layers, quantized to 4-bit precision using NF4 with double quantization (Dettmers et al., 2023). All experiments use bfloat16 compute precision. We chose Gemma-2-9B because it possesses baseline linguistic knowledge of Turkic

languages from pretraining, allowing us to isolate the spectral dynamics of LoRA *adaptation* from primary language acquisition.

3.2 Training Data

We compile monolingual corpora of approximately 150 MB for each of the three target languages:

- **Kazakh (KZ):** 61,879 records from Kazakh Wikipedia and the sozkz-corpus, totaling 14.1M tokens (151.5 MB).
- **Kyrgyz (KY):** 17,360 records from a curated local corpus comprising literature, history, and encyclopedic texts, totaling 4.4M tokens (151.4 MB).¹
- **Uzbek (UZ):** 21,242 records from uzbooks, Uzbek Wikipedia, and FineWeb-2 with Cyrillic script filtering, totaling 5.4M tokens (152.6 MB).

Additionally, we compile an **English (EN)** control corpus of 52,268 records from English Wikipedia (155 MB, 12.7M tokens), matched in size to the Turkic corpora. All texts are chunked into segments of 200–5,000 characters with a 10% held-out validation split.

3.3 LoRA Configuration

We apply LoRA adapters to all attention and MLP projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` (7 modules per layer, 294 total across 42 layers). We set $\alpha = 2r$ in all experiments and use the paged AdamW optimizer with a cosine learning rate schedule and 5% warmup. Table 1 summarizes all experimental configurations.

3.4 SVD Monitoring Protocol

We perform singular value decomposition (SVD) of each `lora_B` weight matrix every 100 training steps, computing five spectral metrics: **Spectral Energy (SE)** $= \sigma_1 / \sum_i \sigma_i$, which quantifies the dominance of the largest singular value; **Effective Rank** (Roy and Vetterli, 2007), measuring the number of significantly contributing singular values; **Stable Rank** $= \|W\|_F^2 / \sigma_1^2$; **SVD Entropy**;

¹Due to licensing restrictions, the Kyrgyz corpus cannot be publicly redistributed. Details are provided in Appendix B.

and **Frobenius Norm** $\|W\|_F = \sqrt{\sum_i \sigma_i^2}$. We monitor both `lora_A` and `lora_B` Frobenius norms separately. The spectral collapse threshold is set at $SE = 0.7$ following Biderman et al. (2024). Full metric definitions are provided in Appendix A.

3.5 Evaluation Protocol

We evaluate each adapter on three tasks: **Perplexity (PPL)** on held-out validation splits (500 samples, 113–128K tokens per language), constructing a *cross-lingual perplexity matrix* by evaluating each adapter on all three languages; **Named Entity Recognition (NER)** using WikiANN (Pan et al., 2017) with 3-shot prompting on 100 test samples per language; and **Question Answering (QA)** using TUMLU (Isbarov et al., 2024), a multiple-choice Turkic language understanding benchmark with 5-shot log-likelihood evaluation.

3.6 Subword Tokenization Analysis

All three Turkic languages exhibit severe subword fragmentation at approximately $3.2\times$ the rate of English (Kazakh: 3.74, Kyrgyz: 3.69, Uzbek: 3.68 tokens/word vs. English: 1.15), reflecting the fundamental mismatch between the Gemma tokenizer and agglutinative Turkic morphology. The implications of this fragmentation for training dynamics are analyzed in Appendix G.

4 Results

4.1 Baseline Performance

Table 2 presents the primary evaluation metrics across all experiments. The Kazakh baseline (E1) achieves the best target-language perplexity (2.73), likely reflecting the larger effective token count (14.1M vs. 4.4–5.4M) and better representation in Gemma’s pretraining data. The Kyrgyz baseline (E3) shows the weakest overall performance (PPL 4.78, NER F1 0.157), consistent with Kyrgyz being the least resourced of the three languages.

The most striking result is the $r=64$ configuration (E5): despite achieving a target-language perplexity of only 5.90, it suffers *complete* NER failure with $F1 = 0$ across all three languages (100% parse failures), indicating total loss of instruction-following capabil-

ID	Experiment	Language	Rank	LR	Epochs	Trainable %	Init
E1	KZ Baseline	Kazakh	16	2×10^{-4}	3	1.05%	Random
E2	UZ Baseline	Uzbek	16	2×10^{-4}	3	1.05%	Random
E3	KY Baseline	Kyrgyz	16	2×10^{-4}	3	1.05%	Random
E4	KY Overfit	Kyrgyz	16	2×10^{-4}	10	1.05%	Random
E5	KY $r=64$	Kyrgyz	64	5×10^{-4}	5	4.08%	Random
E6	KZ→KY Transfer	Kyrgyz	16	2×10^{-4}	3	1.05%	KZ ckpt
E7	Diverged	KY [‡]	64	1×10^{-3}	3	4.08%	Random
E8	EN Control	English	16	2×10^{-4}	3	1.05%	Random

Table 1: Summary of experimental configurations. All experiments use Gemma-2-9B with 4-bit quantization, $\alpha = 2r$, cosine LR schedule, and effective batch size 16. [‡]E7 was trained on Kyrgyz data but diverged at step 800; it is excluded from downstream evaluation but analyzed in Appendix E.

ID	Experiment	Target Lang	PPL (target)	NER F1 (KY)	NER F1 (KZ)	NER F1 (UZ)	TUMLU (target)	TUMLU (KZ)
E1	KZ Baseline	KZ	2.73	0.219	0.164	0.578	38.4%	38.4%
E2	UZ Baseline	UZ	4.03	0.138	0.167	0.287	35.4%	33.1%
E3	KY Baseline	KY	4.78	0.157	0.190	0.326	34.7%	30.2%
E4	KY Overfit (10ep)	KY	4.18	0.173	0.238	0.239	33.9%	27.8%
E5	KY $r=64$	KY	5.90	0.000	0.000	0.000	23.2%	22.2%
E6	KZ→KY	KY	4.73	0.253	0.130	0.348	33.7%	31.7%
E8	EN Control	EN	5.54*	0.209	0.217	0.338	37.7% [†]	36.3%

Table 2: Main evaluation results across all experiments. PPL is computed on the target language validation set. NER F1 is micro-averaged across entity types using 3-shot WikiANN evaluation (100 samples). TUMLU accuracy is reported for the target language and Kazakh (as a forgetting diagnostic). The $r=64$ experiment (E5) shows complete NER failure (100/100 parse failures) despite only moderate PPL degradation. *EN Control PPL is reported on Kazakh. [†]EN Control TUMLU is reported on Kyrgyz.

ity. The TUMLU evaluation corroborates this: accuracy drops to 23.2% (near the 25% random baseline), confirming that the functional collapse extends beyond instruction-following to language understanding itself. We note that Wilson confidence intervals at $n=100$ are wide (e.g., E6 KY NER: 0.253 ± 0.085), so we focus on qualitative distinctions (functional vs. collapsed) and consistent directional patterns across metrics.

4.2 Cross-Lingual Degradation

Figure 1 visualizes the cross-lingual perplexity matrix as a heatmap (full numerical values in Appendix C). The $r=64$ configuration induces catastrophic forgetting: Kazakh PPL rises to 659.25 and Uzbek to 742.73 ($>180\times$ increases). The KZ→KY transfer retains substantially more Kazakh knowledge (KZ PPL 23.49 vs. 47.58 for KY baseline). Extended training (10 epochs, E4) nearly doubles cross-lingual PPL relative to the 3-epoch baseline.

Critically, the **EN control experiment** (E8) reveals that Frobenius norm growth alone does not predict forgetting. Despite exhibit-

ing $33\times$ norm growth—comparable to the KZ baseline ($38\times$)—the EN adapter induces minimal Turkic language degradation (KZ PPL 5.54 vs. 47.58 for the KY baseline). This demonstrates that forgetting severity depends not only on the magnitude of weight perturbations but also on their *alignment* with pre-trained representations.

4.3 Spectral Dynamics: Challenging H1

Our central finding is that the spectral energy threshold of 0.7 proposed by Biderman et al. (2024) is **never reached** in any of our experiments, including those exhibiting severe functional degradation. Figure 2 shows the evolution of mean and maximum SE across training steps.

The KY baseline ($r=16$) shows a mild increase in maximum SE from approximately 0.44 to 0.59, well below the 0.7 threshold. More strikingly, the pathological $r=64$ configuration exhibits *decreasing* SE throughout training, from 0.21 to 0.10–0.13. This counter-intuitive pattern arises because SE is a *shape*

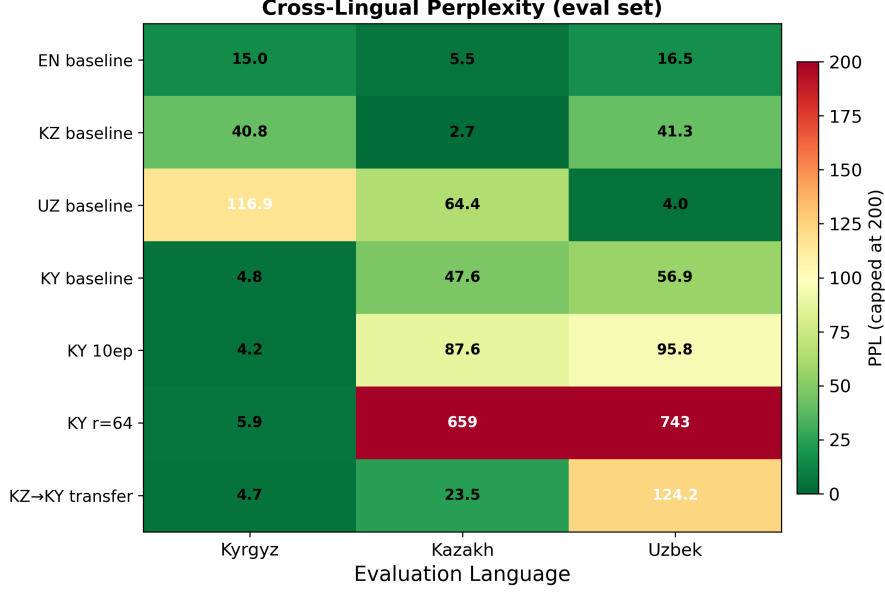


Figure 1: Cross-lingual perplexity heatmap. Darker cells indicate higher perplexity (more forgetting). The $r=64$ row is clearly pathological, while the KZ→KY transfer row shows the most balanced profile.

metric: as training progresses, all singular values grow, but growth is distributed across multiple directions, causing the relative contribution of σ_1 to decrease. The effective rank correspondingly *increases* from approximately 6 to 26 (Appendix E), signaling healthy dynamics under standard interpretations—yet this same configuration suffers complete NER failure.

Figure 3 confirms that no individual layer approaches the collapse threshold. **H1 is not supported in our setting:** SE below 0.7 does *not* guarantee healthy training. The fundamental limitation is that SE is a shape metric invariant to rescaling—a weight matrix with singular values $[100, 50, 25]$ and one with $[0.001, 0.0005, 0.00025]$ have identical SE despite vastly different perturbation magnitudes. For higher-rank configurations, SE is additionally *rank-biased*: since $SE \approx 1/r$ at initialization, higher-rank adapters appear spectrally healthier by definition.

4.4 Frobenius Norm: The Missing Diagnostic

We find that the **Frobenius norm growth ratio** of the `lora_B` matrices provides a clear separation between healthy and pathological training regimes. Table 3 shows that the KZ→KY transfer exhibits only +13% growth, while the pathological $r=64$ configura-

Experiment	$\ \mathbf{B}\ _F^{\text{init}}$	$\ \mathbf{B}\ _F^{\text{final}}$	Growth
KZ→KY Transfer	3.043	3.436	1.13×
KY Baseline $r=16$	0.227	1.899	8.36×
EN Control $r=16$	0.092	3.076	33.4×
KZ Baseline $r=16$	0.081	3.107	38.5×
KY $r=64$	0.566	8.859	15.6×
Diverged $r=64$	—	—	explodes

Table 3: Frobenius norm growth ratios for `lora_B` matrices. EN control and KZ baseline show comparable growth (~ 33 – $38\times$), yet EN produces minimal forgetting while KY baseline with only $8\times$ growth causes severe forgetting—demonstrating that language representation in the pretrained model modulates the impact of weight perturbations.

tion shows 15 – $38\times$ growth. However, the EN control complicates the picture: despite $\sim 33\times$ growth, it causes minimal forgetting, revealing that norm growth is a reliable diagnostic *within* a language category but not *across* categories.

The Frobenius norm captures the *scale* of weight perturbations that SE discards, but the *direction* of those updates—whether they reinforce or disrupt pretrained representations—determines their downstream impact. We propose preliminary growth thresholds ($< 2\times$: healthy; 2 – $10\times$: normal; $> 10\times$: warning) as coarse guidelines, with important caveats discussed in Appendix D.

SVD Dynamics: Baseline vs Aggressive vs Transfer

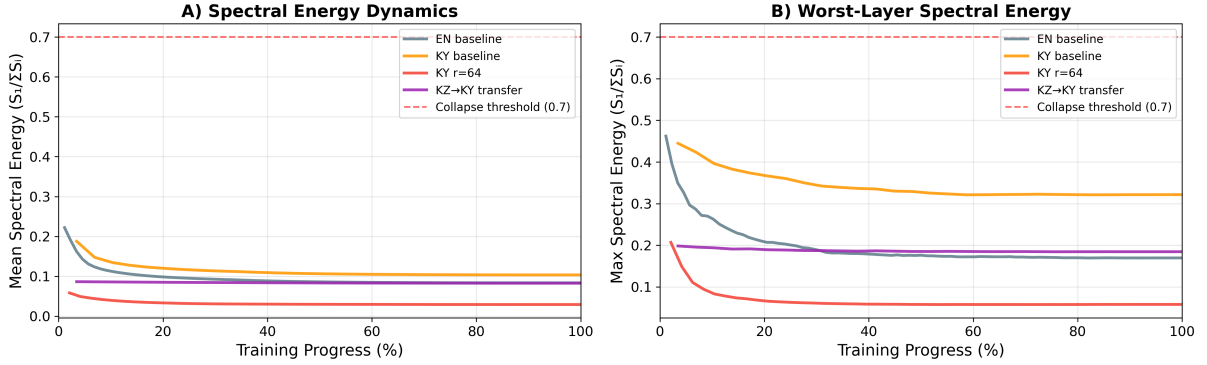


Figure 2: Spectral energy dynamics during training. Top: mean SE across all layers. Bottom: maximum SE across layers. No experiment approaches the proposed 0.7 threshold. The $r=64$ configuration shows *decreasing* SE despite suffering the worst functional degradation.

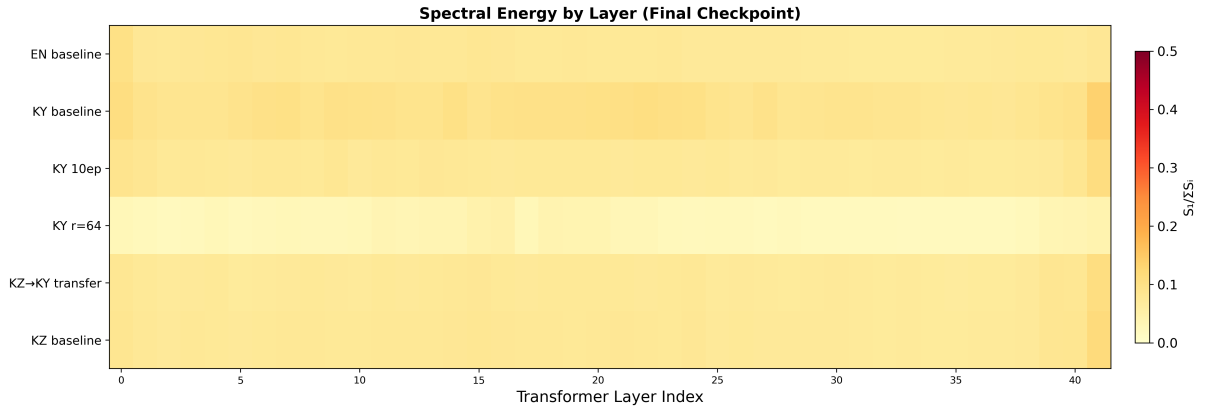


Figure 3: Layer-wise spectral energy at the final training checkpoint. Each row is an experiment; each column is a transformer layer. No layer in any experiment exceeds the 0.7 threshold.

4.5 Cross-Lingual Transfer

The KZ→KY transfer experiment (E6) initializes LoRA adapters from the final Kazakh checkpoint before continuing training on Kyrgyz data. Table 4 shows substantial improvements: Kyrgyz NER F1 improves by +61% (0.253 vs. 0.157), target-language PPL is essentially identical (4.73 vs. 4.78), and Kazakh knowledge is substantially better retained (KZ PPL 23.49 vs. 47.58).

However, Uzbek PPL degrades substantially (124.19 vs. 56.85), suggesting that sequential transfer between closely related languages (Kazakh and Kyrgyz, both Kipchak-branch, Cyrillic-script) can *harm* more distant relatives (Uzbek, Karluk-branch, historically Latin-script).

H2 is partially confirmed: cross-lingual transfer improves downstream performance

and retains source-language knowledge, but the mechanism appears to be *better initialization* rather than *slower degradation*. The transfer adapter begins with an already-trained weight configuration (Frobenius norm growth of only +13%), requiring minimal additional adaptation. Related-language pre-training acts as a *spectral regularizer*: by providing a warm-start with already-structured singular value distributions, it constrains subsequent weight updates to a low-perturbation regime that preserves multilingual knowledge.

5 Analysis and Discussion

Why SE fails. The failure of the SE threshold stems from LoRA initialization: standard LoRA initializes B with zeros, so after the first gradient update B acquires small singular values with $SE \approx 1/r$. As training progresses,

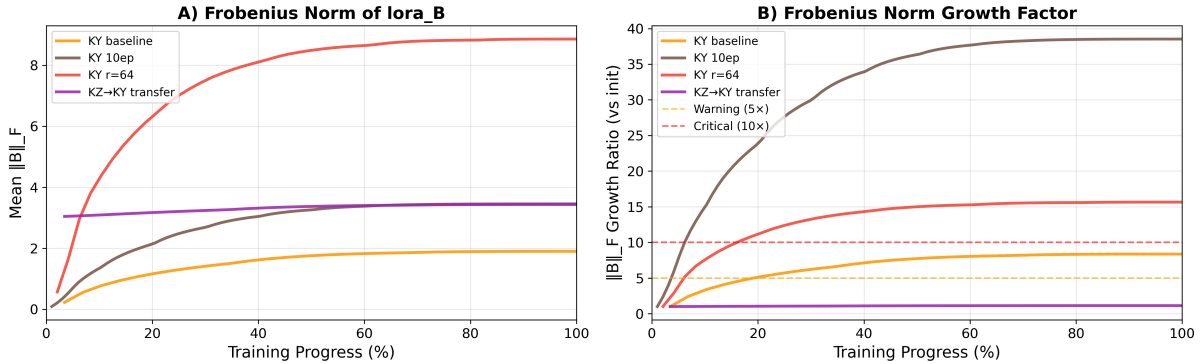


Figure 4: Frobenius norm dynamics of `lora_B` matrices during training. The $r=64$ configuration shows dramatically larger norm growth compared to $r=16$ baselines. The transfer experiment begins at the KZ baseline’s final norm and shows minimal additional growth.

Metric	KY Baseline	KZ→KY
PPL (KY)	4.78	4.73
PPL (KZ)	47.58	23.49
PPL (UZ)	56.85	124.19
NER F1 (KY)	0.157	0.253
NER F1 (KZ)	0.190	0.130
NER F1 (UZ)	0.326	0.348
TUMLU (KY)	34.7%	33.7%
TUMLU (KZ)	30.2%	31.7%
$\ B\ _F$ growth	8.36×	1.13×

Table 4: Comparison of direct Kyrgyz training (E3) and Kazakh→Kyrgyz transfer (E6). Transfer improves KY NER F1 by +61% and retains KZ knowledge (PPL 23.49 vs. 47.58). However, UZ PPL increases (124.19 vs. 56.85), suggesting the benefit is primarily between closely related Kipchak languages.

growth is distributed across multiple directions, causing SE to remain stable or decrease. For $r=64$, the SE denominator includes more singular values, mechanically driving SE lower ($1/64 \approx 0.016$ vs. $1/16 = 0.0625$), making higher-rank adapters appear healthier by definition—a dangerous trap for practitioners monitoring SE across different rank configurations.

Directionality matters. The EN control experiment reveals that the critical factor in forgetting is not the *magnitude* of weight updates but their *directionality*. English fine-tuning reinforces already well-represented features with minimal cross-lingual interference, while Turkic fine-tuning forces the model to restructure underrepresented regions of the

weight space, disrupting shared multilingual features. This is compounded by the $3.2\times$ sub-word fragmentation: Turkic sentences require $\sim 3\times$ more tokens, reducing the information density per gradient update and forcing larger weight modifications (see Appendix G).

The overfit paradox. Extended training (10 epochs, E4) *improves* target-language PPL (4.18 vs. 4.78) despite clear overfitting, but at a steep cost in cross-lingual knowledge (KZ PPL nearly doubles). This suggests a regime of “useful overfitting” where the model memorizes target-language patterns more deeply at the expense of multilingual competence. Full analysis is provided in Appendix F.

6 Limitations

Our study has several important limitations. First, all experiments use a single base model (Gemma-2-9B); results may differ for other architectures. Second, while the EN control uses a matched corpus size, differences in data distribution and tokenizer coverage may partially confound the comparison. Third, NER evaluation relies on few-shot prompting, introducing noise from instruction-following variation. Fourth, we examine only three of ~ 35 Turkic languages, excluding Turkish. Fifth, experiment E5 ($r=64$) simultaneously increases both rank and learning rate relative to baselines, confounding these factors; a controlled $r=64$, $lr=2\times 10^{-4}$ experiment would be needed to isolate rank effects. Sixth, our SVD monitoring focuses on `lora_B` matrices; `lora_A` dynam-

ics may reveal complementary patterns. Seventh, 4-bit quantization may influence spectral properties in ways not captured by our analysis. Finally, our Frobenius norm thresholds are derived from a single experimental setting and should be treated as preliminary guidelines.

7 Conclusion

We have presented a systematic study of the spectral dynamics of LoRA adapters during fine-tuning on three low-resource Turkic languages, yielding three principal findings. First, the spectral energy threshold hypothesis is **not supported**: no experiment exceeds SE of 0.59, yet $r=64$ configurations suffer complete functional collapse. Second, **Frobenius norm growth ratio** is a useful but language-dependent diagnostic—it reliably separates healthy from pathological regimes within underrepresented languages, but well-represented languages absorb comparable growth without degradation. Third, **sequential cross-lingual transfer** between related Turkic languages is highly effective: KZ→KY transfer improves NER F1 by +61% while retaining Kazakh knowledge.

Based on these findings, we recommend that practitioners: (1) monitor Frobenius norm growth rather than spectral energy; (2) avoid aggressive rank-LR combinations for small corpora; (3) use related-language transfer when available; (4) apply cross-lingual perplexity monitoring as a forgetting diagnostic; and (5) consider early stopping by cross-lingual PPL rather than training loss.

Future work should validate these findings across diverse architectures (particularly in full-precision BF16), extend the analysis to additional Turkic languages, and investigate directional norm regularization as a training objective.

Ethics Statement

This work involves fine-tuning language models on text corpora from three Turkic languages and English. While we aim to improve NLP capabilities for underserved language communities, we acknowledge that language models can amplify biases present in training data. Our training corpora, drawn from

Wikipedia, literary texts, and web sources, may contain cultural biases or factual inaccuracies.

Data availability: The Kazakh and Uzbek corpora are compiled from publicly available sources. The Kyrgyz corpus is curated from licensed sources and cannot be redistributed. We release all trained LoRA adapters, SVD monitoring logs, evaluation results, and full training code, enabling complete reproduction of the evaluation and analysis stages.

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LLMs Usage Statement. During the preparation of this manuscript, we made limited use of large language models (LLMs), specifically Anthropic’s Claude, to assist with language refinement and organization of some sections. All technical content, equations, derivations, and experimental design were developed entirely by the authors. The LLM was not used for ideation of methods, data analysis, or generation of results.

Reproducibility Statement

All experimental code (training with SVD monitoring, evaluation, and plotting scripts), fine-tuned LoRA adapters for all eight experiments, SVD monitoring logs, and evaluation reports are publicly available at <https://anonymous.4open.science/r/Spectral-Collapse-Turkic-EMNLP-7D8F>.

The Kazakh, Uzbek, and English training corpora are derived from publicly available sources and can be reconstructed using the provided data preparation scripts. The Kyrgyz corpus cannot be redistributed due to licensing restrictions; however, the released KY adapters and SVD logs enable full reproduction of all evaluation and analysis stages.

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A SVD Metric Definitions

Spectral Energy (SE) quantifies the dominance of the largest singular value:

$$\text{SE} = \frac{\sigma_1}{\sum_{i=1}^r \sigma_i} \quad (1)$$

Effective Rank (Roy and Vetterli, 2007) measures the number of significantly contributing singular values:

$$\text{EffRank} = \exp \left(- \sum_{i=1}^r p_i \log p_i \right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j} \quad (2)$$

Stable Rank provides a continuous relaxation of matrix rank:

$$\text{StableRank} = \frac{\|W\|_F^2}{\sigma_1^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2} \quad (3)$$

SVD Entropy measures the uniformity of the singular value distribution:

$$H_{\text{SVD}} = - \sum_{i=1}^r p_i \log_2 p_i \quad (4)$$

Frobenius Norm captures the overall magnitude of the weight matrix:

$$\|W\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2} \quad (5)$$

B Detailed Training Configurations

All experiments were conducted on a single NVIDIA GeForce RTX 5080 (16 GB) using PyTorch 2.10, Transformers 5.2, and PEFT 0.18. Table 5 provides the full training statistics.

ID	Experiment	Train Loss	Eval Loss	Eval PPL	Time (min)
E1	KZ Baseline	0.275	1.259	3.52	477
E2	UZ Baseline	1.193	1.822	6.18	341
E3	KY Baseline	0.815	1.920	6.82	183
E4	KY Overfit	1.050	1.963	7.12	1193
E5	KY $r=64$	0.274	2.164	8.71	157
E6	KZ→KY	1.809	1.932	6.90	355
E8	EN Control	1.821	2.017	7.51	1351

Table 5: Full training statistics. Eval loss and PPL are computed on the validation split during training. The higher train loss for E6 reflects the adapter starting from KZ-optimized weights.

C Cross-Lingual Perplexity Matrix

Trained on	KY PPL	KZ PPL	UZ PPL
KZ Baseline	40.75	2.73	41.27
UZ Baseline	116.89	64.44	4.03
KY Baseline	4.78	47.58	56.85
KY Overfit	4.18	87.65	95.76
KY $r=64$	5.90	659.25	742.73
KZ→KY	4.73	23.49	124.19
EN Control	14.96	5.54	16.55

Table 6: Cross-lingual perplexity matrix. Each row is an adapter; each column is the evaluation language. Bold indicates target-language PPL. The $r=64$ configuration shows catastrophic cross-lingual degradation ($> 600\times$ PPL increase), while the EN control shows minimal Turkic forgetting despite comparable Frobenius norm growth ($33\times$).

D Frobenius Norm Growth Thresholds

Our experiments suggest the following practical thresholds for $\|B\|_F$ growth monitoring:

- $< 2\times$: **Healthy** training; minimal forgetting (e.g., transfer setup: $1.13\times$).
- $2\text{--}10\times$: **Normal** training; moderate forgetting expected (e.g., KY baseline: $8.36\times$).
- $> 10\times$: **Warning**; significant forgetting likely (e.g., $r=64$: $15\text{--}38\times$).
- Unbounded growth: **Training failure**; divergence imminent (e.g., E7).

These thresholds refer to *relative growth* (ratio of final to initial Frobenius norm). **Important caveats:** The KZ baseline (E1, $38.5\times$) and EN control (E8, $33.4\times$) both exceed the $> 10\times$ warning threshold without functional collapse, while the KY baseline (E3, $8.36\times$) falls within the “normal” range yet shows significant cross-lingual forgetting. We recommend these thresholds as coarse guidelines rather than precise decision boundaries.

E Effective Rank Dynamics

Figure 5 shows the effective rank dynamics across experiments. The $r=64$ configuration shows increasing effective rank (from ~ 6 to ~ 26), indicating growing parameter utilization. The diverged experiment (E7) is the only configuration where effective rank *decreases* (from 43.3 to 37.1), consistent with true spectral collapse.

E7 divergence analysis. The diverged experiment ($r=64$, $\text{lr}=10^{-3}$) provides the most extreme datapoint. Training terminated at step 800 due to loss divergence. Even at this point, maximum SE was only 0.18—still far below the 0.7 threshold—while mean Frobenius norm had already reached 6.75 ($19\times$ growth in just 800 steps). The Frobenius norm shows clear exponential growth from step 100 onward, providing a much earlier warning signal than SE.

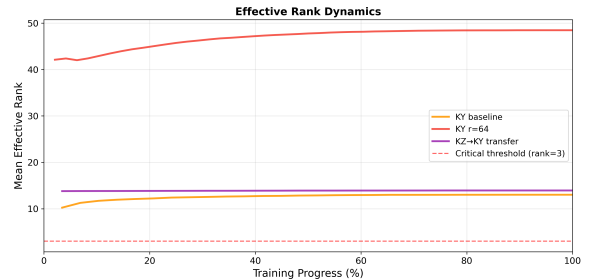


Figure 5: Effective rank dynamics during training. The $r=64$ configuration shows increasing effective rank despite functional degradation, further evidence that standard spectral metrics can be misleading.

F The Overfit Paradox

The 10-epoch Kyrgyz experiment (E4) reveals a paradoxical pattern (Figure 6): extended training *improves* target-language perplexity (4.18 vs. 4.78 for the 3-epoch baseline) despite clear indicators of overfitting (final train loss 0.165 vs. 0.815 for the baseline). However, this improvement comes at a steep cost in cross-lingual knowledge: Kazakh PPL nearly doubles (87.65 vs. 47.58), Uzbek PPL increases by 68% (95.76 vs. 56.85), and TUMLU accuracy on Kazakh drops from 30.2% to 27.8%. The spectral profile remains healthy throughout ($SE < 0.5$), further confirming that SE is not a reliable indicator of training pathology.

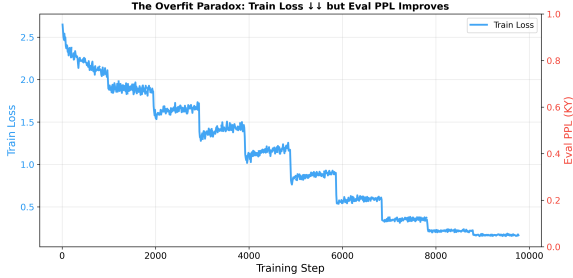


Figure 6: The overfit paradox: 10 epochs of Kyrgyz training (E4) yields better target-language PPL (4.18 vs. 4.78) despite increasing cross-lingual degradation. Train loss continues to decrease while cross-lingual PPL rises monotonically.

G Role of Subword Fragmentation

Language	Tokens/Word	Ratio to EN
English	1.15	1.00×
Uzbek	3.68	3.20×
Kyrgyz	3.69	3.21×
Kazakh	3.74	3.25×

Table 7: Subword fragmentation ratios for the Gemma-2-9B tokenizer.

The $3.2\times$ subword fragmentation ratio has several compounding effects on LoRA adaptation:

Sequence length inflation: A Kyrgyz sentence of 50 words requires approximately $50 \times 3.69 = 185$ tokens, versus $50 \times 1.15 = 58$ tokens for English. With a maximum sequence length of 256 tokens, each training example contains substantially less textual content, reducing the information density of each gradient update.

Morphological fragmentation: Agglutinative suffixes carrying grammatical meaning are split across multiple subwords, forcing the model to reconstruct morphological relationships across token boundaries.

Link to forgetting severity: The EN control provides direct evidence that subword fragmentation interacts with forgetting dynamics. Despite comparable Frobenius norm growth (EN: $33\times$ vs. KZ: $38\times$), the EN adapter causes minimal cross-lingual degradation while Turkic adapters cause severe forgetting, suggesting that the critical factor is the *directionality* of weight updates rather than their magnitude.

H Singular Value Distributions

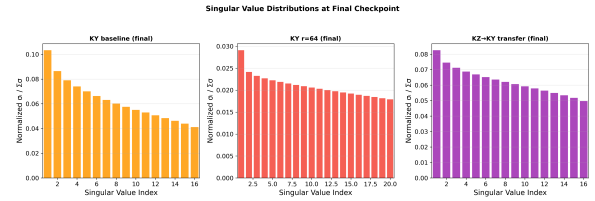


Figure 7: Singular value distributions at the final checkpoint. The $r=64$ configuration (right) shows a well-distributed spectrum (low SE) with dramatically larger absolute magnitudes compared to the $r=16$ baseline (left), explaining why SE fails to detect pathological training.

I Frobenius Norm vs. Cross-Lingual PPL

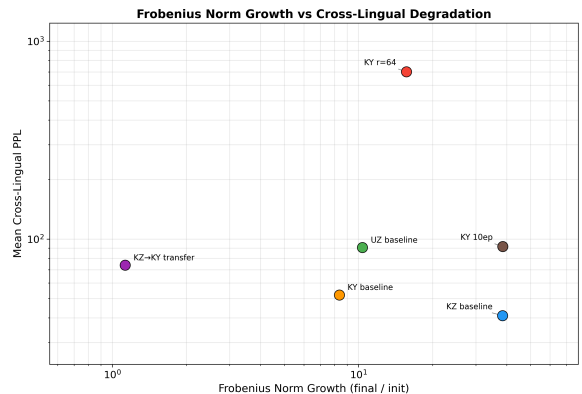


Figure 8: Frobenius norm growth ratio versus cross-lingual perplexity degradation. Within Turkic experiments, the pattern is monotonically consistent ($\rho_s = 0.26$, $n=6$). The EN control is a clear outlier ($33\times$ growth with minimal forgetting), confirming that norm growth is necessary but not sufficient for catastrophic forgetting.

J Detailed Evaluation Results

Table 8 breaks down NER performance by entity type. Table 9 presents TUMLU accuracy across all languages.

Experiment	PER F1	ORG F1	LOC F1
KZ Baseline (on KY)	0.236	0.162	0.229
UZ Baseline (on KY)	0.095	0.150	0.209
KY Baseline	0.101	0.160	0.266
KY Overfit	0.224	0.255	0.119
KY $r=64$	0.000	0.000	0.000
KZ→KY	0.344	0.202	0.214

Table 8: Per-type NER F1 scores on Kyrgyz WikiANN. The KZ→KY transfer achieves the highest PER F1 (0.344). The $r=64$ configuration produces zero scores due to 100% parse failures.

Experiment	KY	KZ	UZ
KZ Baseline	35.7%	38.4%	32.8%
UZ Baseline	31.9%	33.1%	35.4%
KY Baseline	34.7%	30.2%	29.6%
KY Overfit	33.9%	27.8%	31.5%
KY $r=64$	23.2%	22.2%	22.9%
KZ→KY	33.7%	31.7%	30.8%

Table 9: TUMLU accuracy across all experiments and languages. The $r=64$ configuration drops to near-random levels ($\sim 23\%$) across all languages.

Functional collapse examples. To illustrate functional collapse at $r=64$, we present raw model outputs. The model ignores instructions and exhibits degenerate repetition.

NER prompt: Extract named entities from the text. Format each as: ENTITY ||| TYPE

Input: Бишкекте Кыргыз Республикасынын Президентини Садыр Жапаров менен Казакстандын Президентини Касым-Жомарт Токаев жолукту.

$r=64$ **output:** Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон... (repeats indefinitely)

Text continuation prompt: Кыргызстан --- это

$r=64$ **output:** не только, не только, не только, не только, не только... (single bigram repeated 25+ times until max tokens)

Both examples demonstrate complete loss of instruction-following capability—degenerate repetitions are a hallmark of functional collapse distinct from spectral collapse.